# Materials Science
# M.Sc. education program

crei.skoltech.ru/cee/education

msc.skoltech.ru/materials-science

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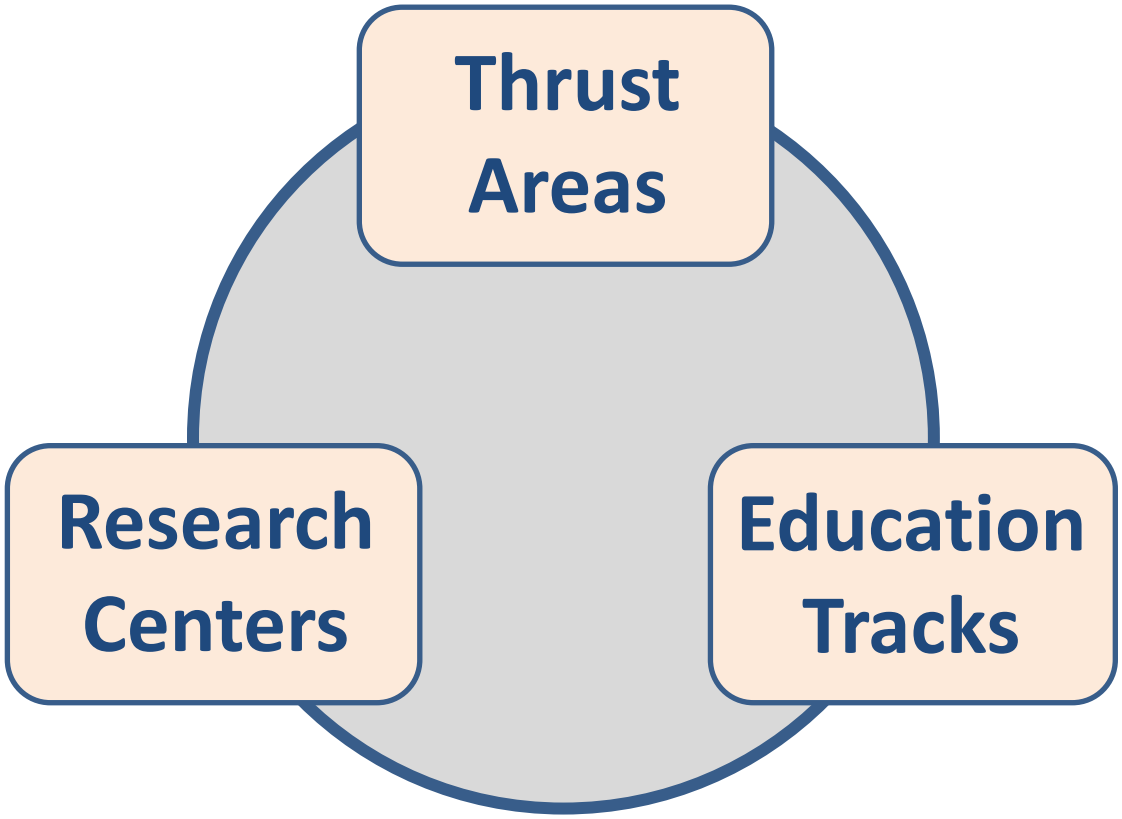
Skoltech
Skolkovo Institute of Science and Technology

# Skoltech research & education environment

- No separate theory or experiment departments
- No separate physics or chemistry departments



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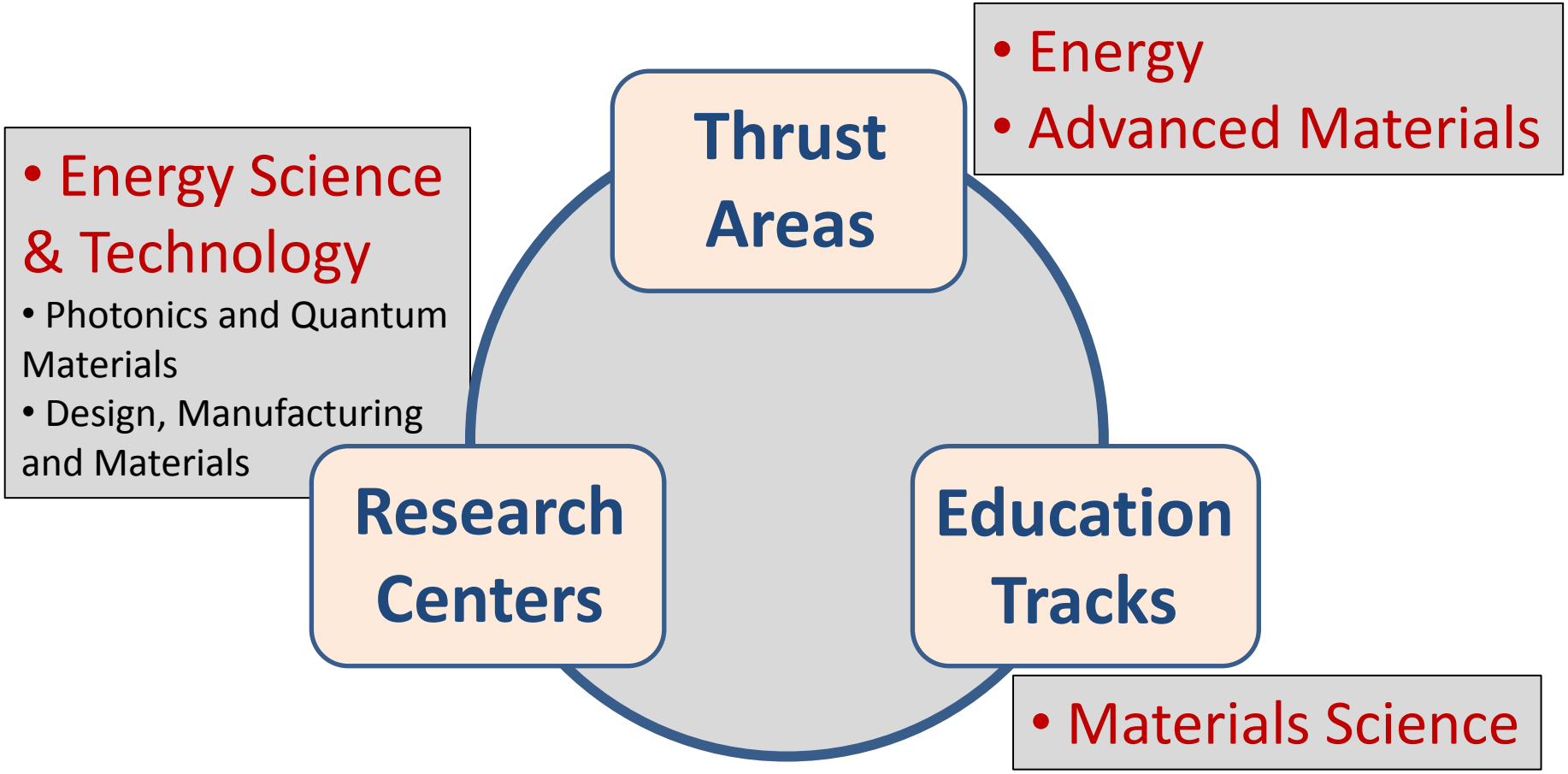
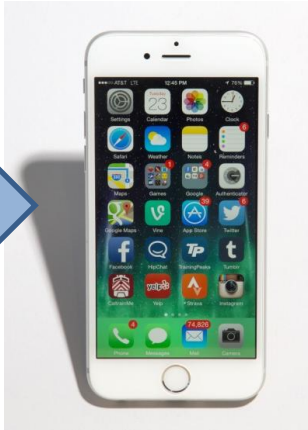


# What is (modern) Materials Science?

*Would not exist without recent advances in Materials Science*



## Elementary devices/parts

- box/frame
- processor (transistors)
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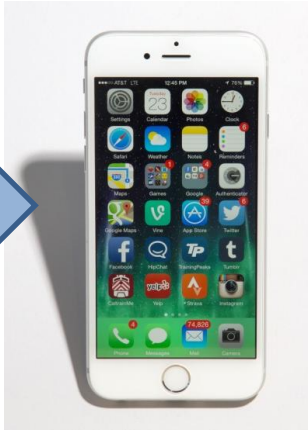

## Elementary devices/parts

- box/frame
- processor (transistors)
- display
- battery
- thermal management
- wires, antennas, audio
- …

## Functional materials

- mechanical & chemical
- semiconductors
- ???

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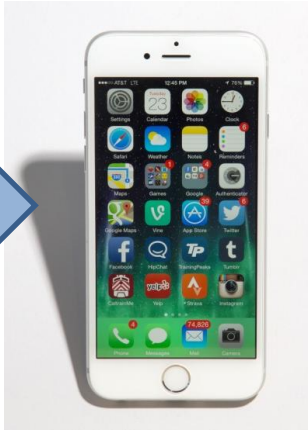

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## Functional materials

- mechanical & chemical
- semiconductors
- light emitter
- battery electrodes
- thermal conductors
- thermal insulators
- …

# What do Materials Scientists do?

OLED vs LCD visualization

Above: Simulation of the difference in contrast from the AC7 OLED to a traditional LCD.

- Create new materials
- Optimize existing materials
- Create/study/optimize elementary devices (lab prototypes)

# Materials Science track: courses

***Core courses***

- Survey of Materials **(Term 1B)**
- Materials Chemistry **(Term 2)**

***Experimental Materials Science***

***Recommended courses***

- Introduction to Solid State Physics**(2)**
- Material Structure Characterization Methods **(3)**
- Organic Materials for Electronics, Energy Conversion and Storage **(3)**
- Electrochemistry **(4)**
- Carbon Nanomaterials **(4)**

***Computational Materials Science***

***Recommended courses***

- Computational Chemistry and Materials Modeling **(2)**
- Structure and Property of Materials **(3)**
- Fundamentals of Device Physics **(4)**
- Advanced Solid State Physics **(6)**

***Elective courses***

- discuss with your advisor

# Career paths

Training time

- (PhD+) **Research & teaching:**
  - Academic universities (e.g. Skoltech)
- (PhD) **Research:**
  - Universities (Skoltech, MIT)
  - Research institutions (Los Alamos Nat Lab)
  - R&D labs in large corporations (Samsung)
  - other high-tech companies
- (MSc) **Engineering, manufacturing, support etc.:**
  - Startups (batteries, solar cells, electronics)
  - Industry (energy, electronics, chemical)

*Skoltech is the right place to start your career of a Materials Scientist or Engineer*

# Job market: Examples from high-tech industry

**Advanced materials**

- Boeing, Mitsubishi
- IMEC

**Energy storage**

- Tesla Motors
- Liotech

**Energy conversion**

- Heliatek, Solarmer

**Electronics**

- Samsung, IBM
- Plastic Logic

**Chemical**

- BASF, Chevron, DuPont

**Automotive**

- Nissan, KAMAZ

**IT (materials modeling)**

- Continuum Analytics, Gaussian, MedeA, Simbeyond
- Kintech (Skolkovo resident)

*Skoltech will make you competitive on this market*

# RESEARCH DOMAIN

Changing the world through creative research

Future IT
Advanced Devices
New Materials
Research Paper
Research Infra

## New Materials

Cutting-edge new materials is the critical factor of both the current as well future IT industries.

Research on new material enabling new display formats, such as reflective, transparent, and/or flexible displays and high-performance optical films and color-converting organics and inorganics are examples of activities necessary for such paradigm shifts.

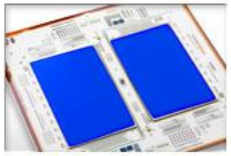
In addition to active electronics materials, SAIT is conducting research on new materials and processing technologies to foster High-performance yet environmentally friendly electronics technologies. Nanomaterial technologies offer an opportunity to reduce consumption of natural resources and energy in manufacturing as well as in the use of electronic devices.

Synthesis and incorporation into devices of graphene, quantum dots are examples of activities that SAIT is pursuing, based on our anticipation of a better future based on technology.

### Organic Emitting Materials

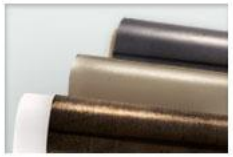
OLED (Organic Light Emitting Diodes) is ... More

### Flexible Electronics



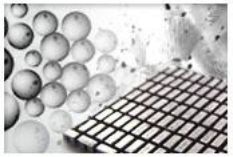
The advent of the so-called "ubiquitous ... More

### Advanced Optical Film



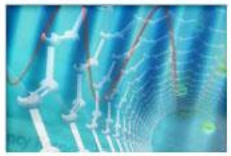
Advanced optical film technology ... More

### Functional Inorganics



Functional inorganics technology ... More

### Nano Carbon Composite



### Quantum Dot

### Next-generation Battery

### Energy Harvesting

# Job market: Established partners for CEE CREI

- Multiple joint projects with MIT, MSU, MIPT

**Joint Labs**

- Joint Lab with MSU "Advanced Materials for Green Energy"
- Joint Lab with Institute of Problems of Chemical Physics
- Joint Lab with Mendeleev University

**Academic – research and internships**

- Los Alamos National Lab, University of Texas at Austin, Nanjing University, Johannes Kepler University/Linz Institute for Organic Solar Cells

**Academic – research**

- University of Bayreuth (Volkswagen), University of Antwerp (research contract), University of Ulm (Helmholtz Institute), University at Albany (CRDF)
- EU framework project COST Action 1307 StableNextSol

**Industry**

- Systems for Microscopy and Analysis (memorandum, two research contracts)
- KAMAZ, SMA, TAGRAS, Nissan, TEEMP, Samsung, Bosch, InEnergy

**Small companies**

- SuperOx, Skolkovo startups

*At Skoltech you will have access to partners via Research & Industrial "immersion"*

# M.Sc. Research in Materials Science

## In brief

- *Highly interdisciplinary*
- Materials for energy storage – Abakumov, Antipov, Stevenson
- Optoelectronic materials – Nasibulin, Troshin
- Structural materials – Korsunsky
- Computational Materials Science – Oganov, Levchenko, Perebeinos, Shapeev, Tretiak, Zhugayevych
- Theoretical Materials Science – Buchachenko, Fine
- *… with details on faculty web-pages: example*

## M.Sc. Thesis 2016-2018

- Materials for energy conversion and storage (11 students)
- Materials for optoelectronics (3 students)
- Other materials (2 students)

*At Skoltech you will do world-level research at cutting edge of science*

# Laboratories in Materials Science

## Experiment (visit Lab Tours)

- Multifunctional CEE CREI Lab
- Laboratory of Nanomaterials (Nasibulin)
- partners (MSU, MIPT, IPCP, MendeleevU)

## Theory (software+hardware)

- Multifunctional CEE CREI Lab
- USPEX Lab (Oganov, computational materials discovery)
- partners (MSU, LANL)

## Main Campus Labs (2019) and Skoltech Shared Facilities

*At Skoltech you will have access to unique labs and facilities*

# What is special about Materials Science at Skoltech?

## 1) Broad coverage of topics

- 12 special courses
- Several state of the art Labs

## 2) Distinguished faculty – median h-index > 35

- Cutting edge of science (worldwide)
- Great presenters (worldwide)

## 3) General Skoltech excellence

- Broad selection of elective courses
- Entrepreneurship and innovation support
- Industrial immersion and Academic mobility
- Modern education – interactive learning

# Welcome to Skoltech

*Further information:*

- Lab Tour
- crei.skoltech.ru/cee/education
- msc.skoltech.ru/materials-science
- If you are interested to join a specific research project contact our faculty